Ryerson University

**Department of Electrical, Computer, & Biomedical Engineering**
Faculty of Engineering & Architectural Science

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| **Course Title** | **ELE827** |
| **Course Name** | **CMOS Mixed Signal** |
| **Semester/Year** | **F2026** |

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| **Instructor** | **Dr Fei Yuan** |

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| **Lab Number** | **Lab 2** |
| **Lab Name** | **Sample and Hold** |

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|---|---|
| **Submission Date** | |
| **Due Date** | |

| Student Name | Student Number | Section | Student Signature |
|---|---|---|---|
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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab reports your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

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Table 1: Component Values

**List of Abbreviations**

ADE — Analog Design Environment
ADE XL — Analog Design Environment Explorer
TB — Testbench
TRAN — Transient Analysis
DC — Direct Current (DC analysis / operating point)
PVT — Process / Voltage / Temperature
TT / FF / SS — Typical / Fast / Slow process corners
VDD — Supply Voltage
GND — Ground
VIN — Input Voltage
VOUT — Output Voltage

## Schematic and Symbols

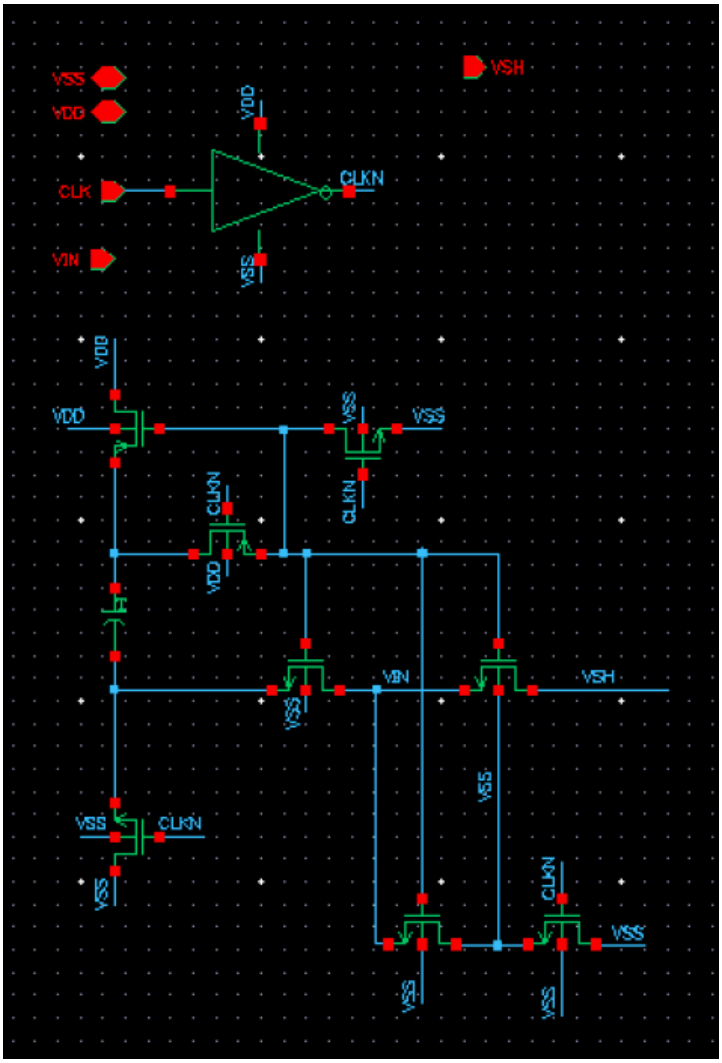



**Figure 1:** Sample and Hold with Bootstrapping

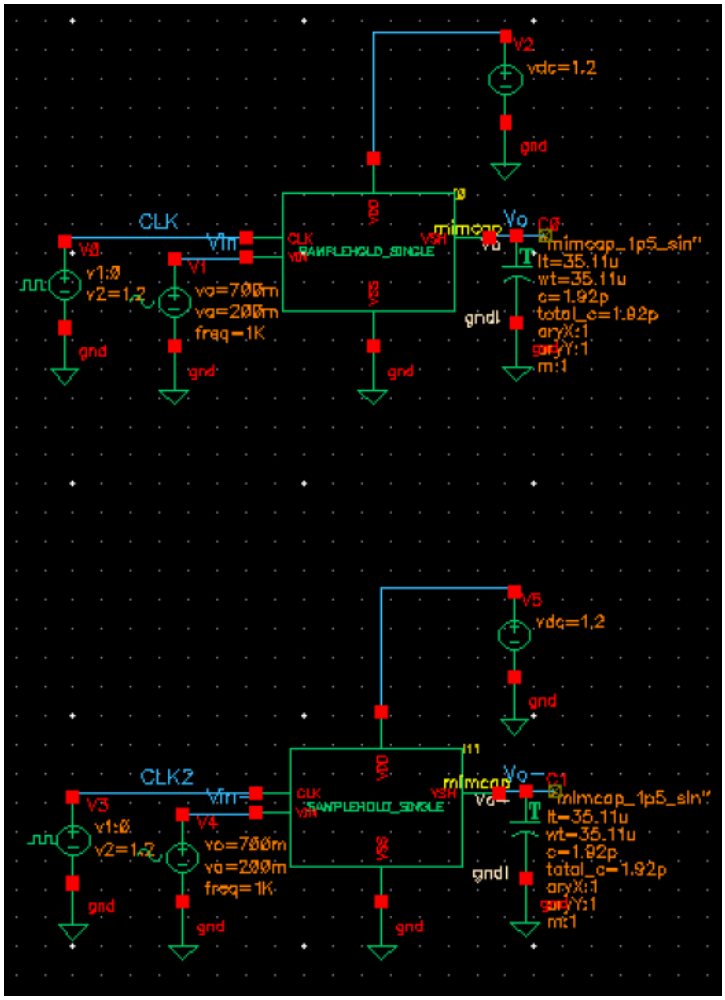



**Figure 2**: Differential Sample and Hold

| Component | Parameter | Value |
|---|---|---|
| **Bootstrap Capacitor** | Capacitance | 2 pF |
| | Dimensions | 35.85 µm × 35.85 µm |
| **M1 Transistor** | Width | 500 nm |

| | | |
|---|---|---|
| | Fingers | 1 |
| **DAC Capacitor** | Capacitance | 1.92 pF |

Table 1: Component Values

## Results and Observations

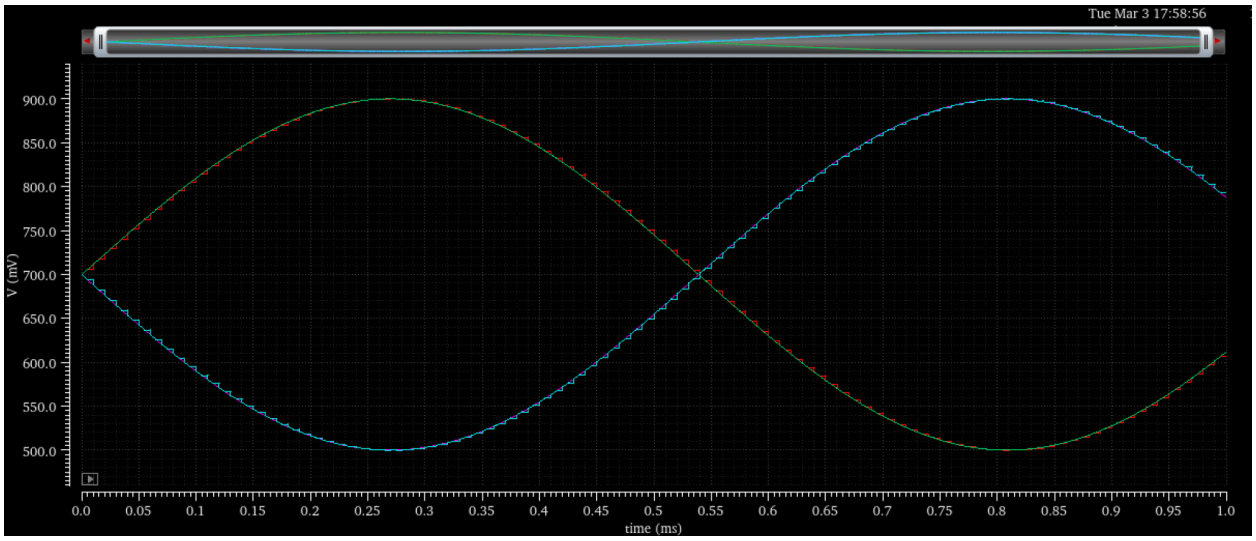



**Figure 3: Output of each individual Sample and Hold with input**

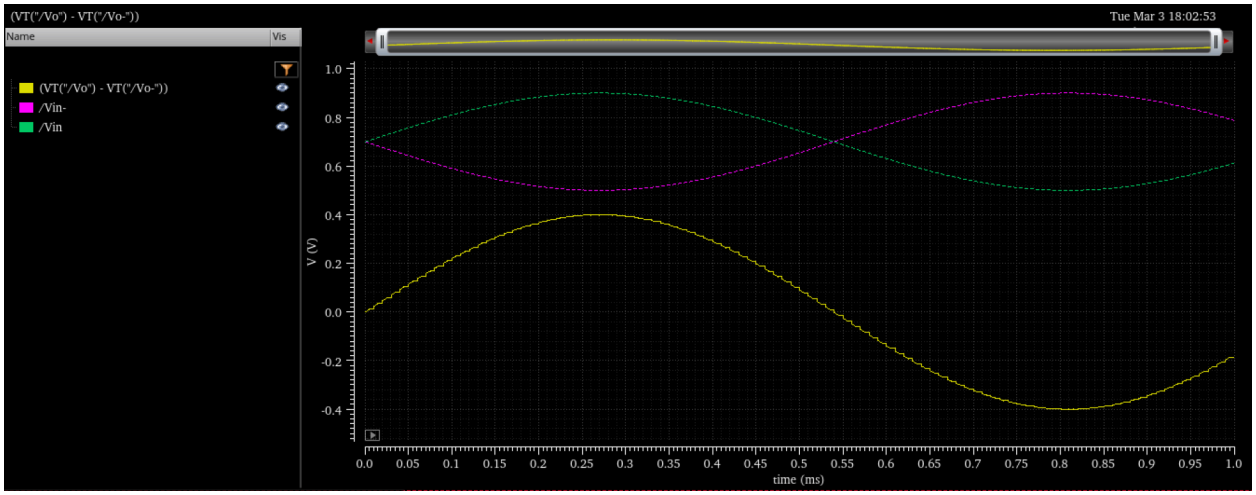



**Figure 4:** Differential Sample and Hold output with both inputs

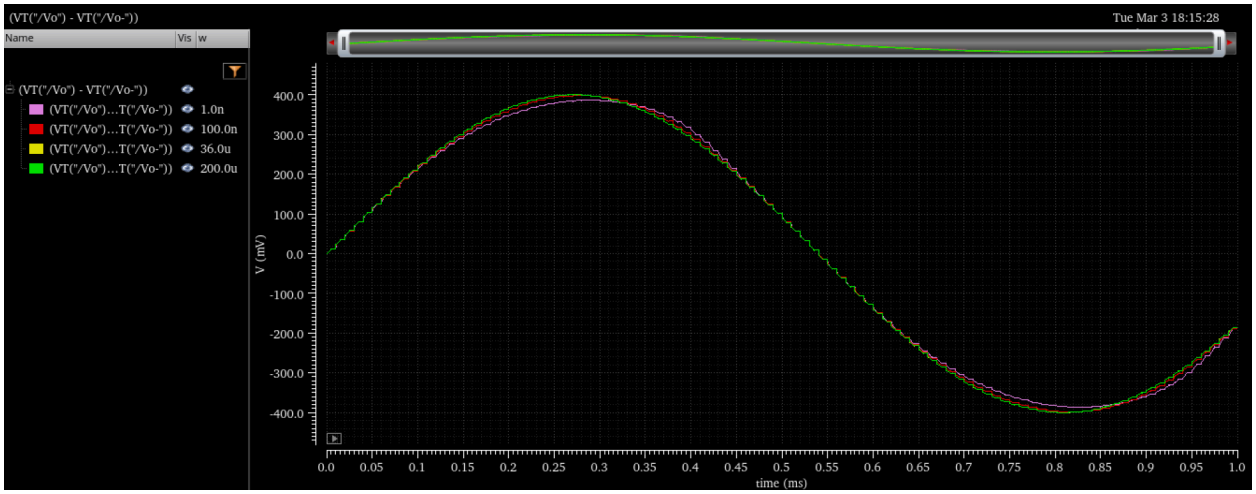



**Figure 5:** Differential Sample and Hold output with varying bootstrap capacitor width

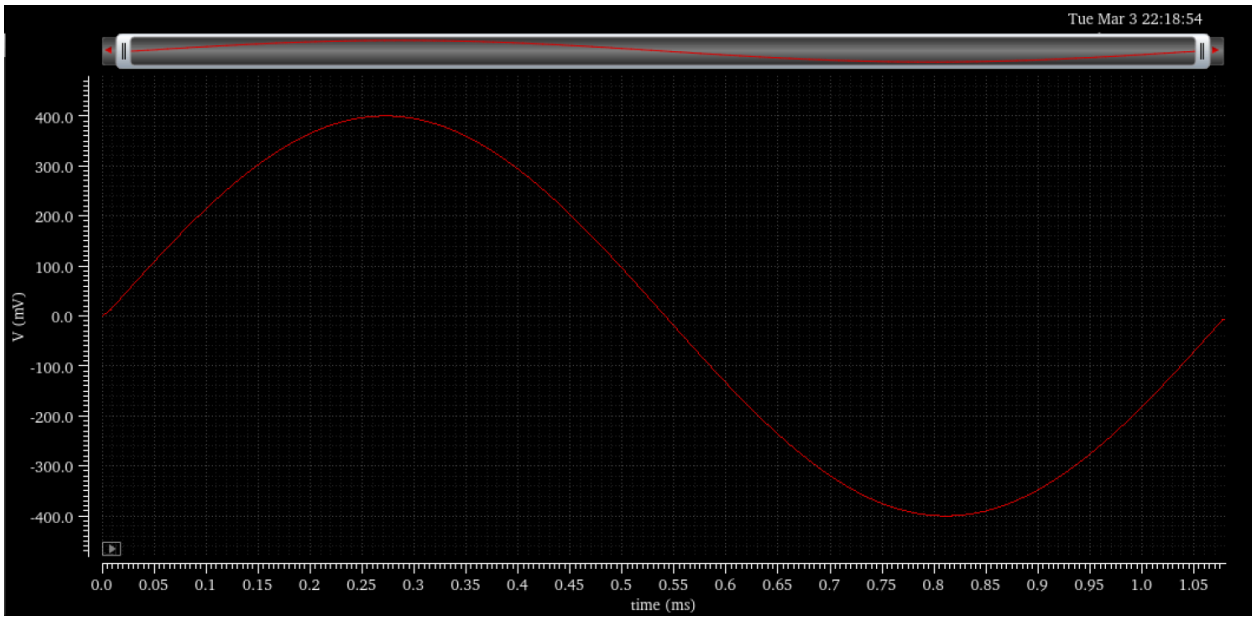



**Figure 6:** Strobed differential output

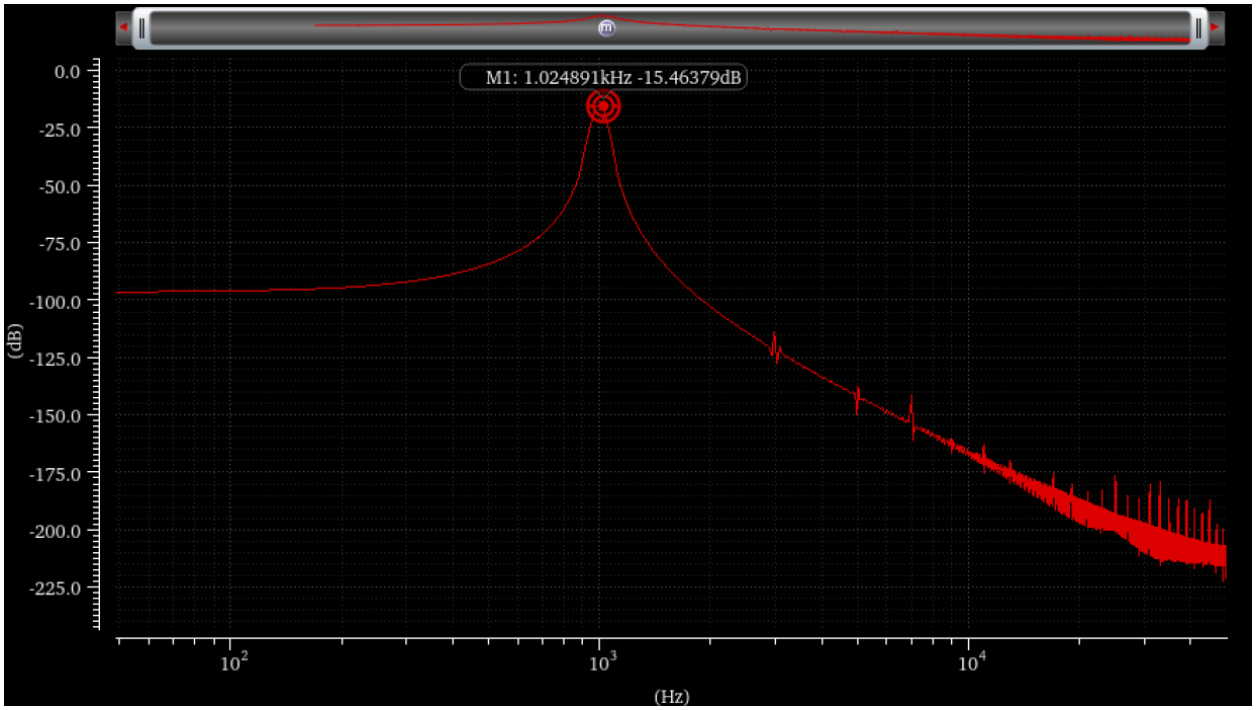



**Figure 7**: Spectrum of Differential Sample and Hold output in log scale

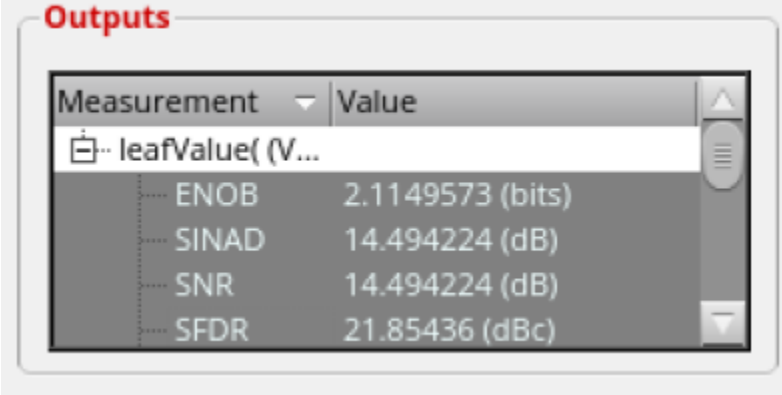



**Figure 8:** SFDR, SNR, SNDR, and ENOB

| | | |
|---|---|---|
| ELE827:SAMPLEHOLD_SINGLE_TEST:1 | average(getData... | 286.9u |

**Figure 9**: AVG Power Calculation

## Conclusion

In this lab, a differential sample hold is designed by combining two bootstrapped samples and holds with inputs varying by 180 degrees. The bootstrapped circuit allows the circuit to M1 transistor to charge faster during the hold to tracking process by allowing a voltage higher than VDD to be applied on the gate of the transistor, allowing a bigger inversion channel thus faster following speed.

It can be seen that from **Figure 5**, increasing the bootstrap capacitors width past a certain point will have no effect on the transient response of the sample and hold. However, at lower widths, the sample and hold can have slightly delayed responses near the peak of the signal.

The differential sample and hold is used in this experiment because it reduces the effect of the intrinsic capacitance of the M1 transistor. This parasitic capacitance can interact with the $C_{DAC}$ and reduce the accuracy of the sampled signal.

In conclusion, the differential bootstrapped sample and hold allows the sampling transistor to track the input signal more quickly by applying a boosted gate voltage, improving the speed of the tracking process. Additionally, the differential circuit reduces the effect of the intrinsic capacitance of the M1 transistor on the $C_{DAC}$, and the results show that increasing the bootstrap capacitor width beyond a certain point does not significantly improve the transient response of the sample and hold.